Scientific Computing Lab 5

Objectives: After completing this lab, students should be able to:

- Apply the finite difference method to discretize and solve one-dimensional heat and wave equations.
- Develop MATLAB programs to compute numerical solutions of time-dependent partial differential equations.
- Analyze the effect of spatial and temporal step sizes on the stability and accuracy of the numerical solution.
- Visualize and interpret numerical results of heat conduction and wave propagation using MATLAB plots.

One-Dimensional Heat Equation

Consider the one-dimensional heat conduction equation

$$C\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \qquad \text{---------- (1)}$$

where $u(x,t)$ - temperature at position x and time t, C- constant (thermal parameter, related to thermal properties of the material), x - spatial variable, t - time variable, $\frac{\partial u}{\partial t}$ - rate of change of temperature with respect to time, $\frac{\partial^2 u}{\partial x^2}$ - spatial curvature of the temperature distribution. This equation describes heat diffusion along a rod.

Discretization of the Domain

The (x,t) plane is divided into rectangular mesh points. Spatial grid: $x = ih, i = 0,1,2, \ldots$ Time grid: $t = jk, j = 0,1,2, \ldots$where h - space step size, k - time step size. Let $u(i,j) = u(x_i, t_j)$ represent the approximate solution at grid point $(x_i\,, t_j)$.

Finite Difference Approximations

Using forward difference approximation

$$\frac{\partial u}{\partial t} \approx \frac{u_{i,j+1} - u_{i,j}}{k}$$

Using central difference approximation

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{1}{h^2}\left(u_{i-1,j} - 2u_{i,j} + u_{i+1,j}\right)$$

Substituting the above approximations into the heat equation:

$$C\frac{u_{i,j+1} - u_{i,j}}{k} = \frac{1}{h^2}(u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

Let $\lambda = \frac{k}{Ch^2}$ Then $u_{i,j+1} = u_{i,j} + \lambda(u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$

or $$\boldsymbol{u_{i,j+1} = (1 - 2\lambda)u_{i,j} + \lambda(u_{i-1,j} + u_{i+1,j}).}$$

This is the **explicit finite difference scheme** for the heat equation.

Note: For the stability of the explicit method,

$$\boldsymbol{\lambda \leq \frac{1}{2}} \qquad \text{(i.e. } \frac{k}{Ch^2} \leq \frac{1}{2}\text{)}$$

Algorithm: Heat Equation (Explicit Scheme)

Input: rod length L, total time T, spatial divisions N, time divisions M, constant C

Step 1: Compute step sizes $h = \frac{L}{N}$, $k = \frac{T}{M}$

Step 2: Compute $\lambda = \frac{k}{Ch^2}$ Check stability condition.

Step 3: Generate spatial grid $x_i = ih$

Step 4: Initialize temperature using initial condition $u_{i,0} = f(x_i)$

Step 5: Apply boundary conditions $u_{0,j} = g_1(t_j)$

$u_{N,j} = g_2(t_j)$

Step 6: For each time level,

For $j = 0,1,2, \dots, M - 1$

Compute interior nodes

$u_{i,j+1} = (1 - 2\lambda)u_{i,j} + \lambda(u_{i-1,j} + u_{i+1,j})$ for $i = 1,2, \dots, N - 1$

Step 7: Repeat until final time level is reached.

Step 8: Display or plot the temperature distribution.

MATLAB CODE

```
% Heat Equation using Finite Difference Method (Explicit Scheme)
clear
clc

L = 1;
T = 0.5;
C = 1;

N = 10;
M = 200;

h = L/N;
k = T/M;

lambda = k/(C*h^2);
```

```
x = 0:h:L;

u = zeros(N+1,M+1);

% Initial condition
for i=1:N+1
   u(i,1) = sin(pi*x(i));
end

% Boundary conditions
u(1,:) = 0;
u(N+1,:) = 0;

% Finite difference iteration
for j=1:M
   for i=2:N
      u(i,j+1) = (1-2*lambda)*u(i,j) + ...
                 lambda*(u(i-1,j) + u(i+1,j));
   end
end

surf(u)
xlabel('Time step')
ylabel('Space node')
zlabel('Temperature')
title('Heat Equation Solution')
```

Example: Consider the heat conduction equation

$$C\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

with boundary conditions $u(0,t) = 0, u(1,t) = 0$. Initial condition $u(x,0) = x(1-x)$.

Take $h = 0.25, k = 0.02, C = 1$.

Grid Points: Spatial nodes

$$x_i = ih$$
$$x = 0,\ 0.25,\ 0.5,\ 0.75,\ 1$$

Time levels $t_j = jk$

Compute λ: $\lambda = \frac{0.02}{(0.25)^2} = 0.32$

Initial Values (*j*= 0)

Using $u(x,0) = x(1-x)$

$$u_{0,0} = 0, u_{1,0} = 0.1875, u_{2,0} = 0.25, u_{3,0} = 0.1875, u_{4,0} = 0$$

i	x_i	$u_{i,0}$	$u_{i,1}$
0	**0**	**0**	**0**
1	**0.25**	**0.1875**	**0.1475**
2	**0.5**	**0.25**	**0.21**
3	**0.75**	**0.1875**	**0.1475**
4	**1**	**0**	**0**

Exercise:

1. Solve the heat equation with

$$u(0,t) = 0, u(1,t) = 0$$
$$u(x,0) = \sin(\pi x)$$

Use $h = 0.25, k = 0.01$. Compute the numerical solution at the first two time levels. Compare the result with the exact solution: $u(x,t) = e^{-\pi^2 t}\sin(\pi x)$

One-Dimensional Wave Equation

Consider the one-dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

where $u(x,t)$ - displacement of the vibrating string at position x and time t, c - wave velocity (a constant), x - spatial coordinate, t – time. This equation describes the vibration of a stretched string.

Finite Difference Approximations

Using central difference approximation

$$\frac{\partial^2 u}{\partial t^2} \approx \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{k^2}$$

Using central difference approximation

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{1}{h^2}(u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

Substituting these approximations into the wave equation gives

$$\frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{k^2} = c^2 \frac{1}{h^2}(u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

Let

$$\lambda = \frac{ck}{h}$$

Then

$$u_{i,j+1} = 2u_{i,j} - u_{i,j-1} + \lambda^2(u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

or

$$\boldsymbol{u_{i,j+1} = -u_{i,j-1} + 2(1 - \lambda^2)u_{i,j} + \lambda^2(u_{i-1,j} + u_{i+1,j})}$$

This is the finite difference scheme for the wave equation.

Note: For numerical stability

$$\lambda \leq 1 \quad (\text{i.e. } \frac{ck}{h} \leq 1)$$

Algorithm

Input: rod length L, total time T, number of spatial divisions N, number of time steps M, wave speed c.

Step 1: Compute step sizes $h = L/N$, $k = T/M$

Step 2: Compute $\lambda = ck/h$. Check stability condition.

Step 3: Generate spatial grid $x_i = ih$

Step 4: Apply initial displacement $u_{i,0} = f(x_i)$

Step 5: Apply initial velocity

$$\frac{\partial u}{\partial t}(x, 0) = g(x)$$

Compute first time level

$$u_{i,1} = u_{i,0} + kg(x_i) + \frac{\lambda^2}{2}(u_{i-1,0} - 2u_{i,0} + u_{i+1,0})$$

Step 6: Apply boundary conditions $u_{0,j} = 0$, $u_{N,j} = 0$

Step 7: For each time level

For $j = 1,2, \dots, M - 1$

Compute

$$u_{i,j+1} = 2u_{i,j} - u_{i,j-1} + \lambda^2(u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

for $i = 1,2, \dots, N-1$

Step 8: Repeat until final time level.

MATLAB CODE

```
% Wave Equation using Finite Difference Method
clear
clc

L = 1;
T = 1;
c = 1;

N = 20;
M = 200;

h = L/N;
k = T/M;

lambda = c*k/h;

x = 0:h:L;

u = zeros(N+1,M+1);

% Initial displacement
for i=1:N+1
   u(i,1) = sin(pi*x(i));
end

% Boundary conditions
u(1,:) = 0;
u(N+1,:) = 0;

% First time step
for i=2:N
   u(i,2) = u(i,1) + 0.5*lambda^2*(u(i+1,1)-2*u(i,1)+u(i-1,1));
end

% Finite difference iteration
for j=2:M
   for i=2:N
```

```
        u(i,j+1) = 2*u(i,j) - u(i,j-1) + ...
                lambda^2*(u(i+1,j)-2*u(i,j)+u(i-1,j));
    end
end

surf(u)
xlabel('Time step')
ylabel('Space node')
zlabel('Displacement')
title('Wave Equation Solution')
```

Example

Solve $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ for $0 < x < 1$ with initial displacement $u(x,0) = \sin(\pi x)$, initial velocity $u_t(x,0) = 0$, boundary conditions $u(0,t) = 0, u(1,t) = 0$. Take $h = 0.1$

Solution: Using the finite difference scheme

$$u_{i,j+1} = 2u_{i,j} - u_{i,j-1} + \lambda^2(u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

The analytical solution is $u(x,t) = \sin(\pi x)\cos(\pi t)$.

x	$t = 0$	$t = 0.25$	$t = 0.5$
0	0	0	0
0.25	0.707	0.5	0
0.5	1	0.707	0
0.75	0.707	0.5	0
1	0	0	0

Exercise:

1. Solve the wave equation $\frac{\partial^2 u}{\partial t^2} = 4\frac{\partial^2 u}{\partial x^2}$ for $0 < x < 1$with $u(x,0) = x(1-x), u_t(x,0) = 0$ and boundary conditions $u(0,t) = 0, u(1,t) = 0$. Compute the numerical solution using the finite difference scheme.